MIT 805

Group Project: Part 1

Mokgopo L u14098718

Dlamini M u18276319

Department of Computer Science

University of Pretoria

September 2026

Dataset Provenance and Description

We picked a dataset from the transport category, particularly the New York City Taxi and Limousine Commission (NYC TLC) Yellow Taxi Trip Records. The dataset is available via the NYC TLC Trip Record Data website: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page. The data is records of trips made by licensed yellow taxi drivers in New York City (NYC). These service providers are in charge of the taxi metering systems in the taxis that collect the records, as such NYC TLC publish the data, but they do not create the records themselves. The dataset is fit for academic use because it is publicly available and released by a New York City government agency for transport research and innovation purposes under the NYC's terms of use. The dataset doesn't contain PI such as names, cell phone number, and exact home addresses.

Dataset Size and characteristics

The TLC Trip Record Data - Yellow Taxi dataset is a large-scale collection of taxi trip records spanning 2009 to May 2026. It is published in Parquet format and contains approximately 1.85 billion rows across 20 variables. The raw dataset size is 26.96 GB compressed, representing the files as stored on the disk. When they are loaded into Spark, the working dataset expands to 17.89 GB in memory, reflecting the decompressed structure Spark uses for its computation. After the schema normalisation and cleaning (removing null timestamps, negative fares, duplicates, and extreme outliers), the processing dataset is reduced to 7.15 GB expanded, which ensures quality and consistency for analysis.

Data Quality and Preparation

To handle schema drift across the monthly Parquet files, schemas were normalized by casting IDs like vendorID to integers and renaming inconsistent columns. Rows with corrupted or null timestamps, drop-off times earlier than pick-up times, and negative fare amounts were removed. Duplicates were dropped, and data was restricted to valid years (2022–2025) to exclude legacy system clock errors. As noted by the NYC TLC, data correctness is not verified by technology providers. Key metrics like passenger counts are driver-reported and may be inaccurate. The tip_amount variable only reflects credit card transactions, and doesn't record cash tip which introduces a structural understatement bias. Outliers (e.g., trips >100 miles) were filtered out to maintain geographical representativeness, though insights remain strictly limited to NYC taxi dynamics. Due to the large data volume, analytical overhead was optimized by selecting only critical columns rather than processing the entire schema footprint.

Exploratory Data analysis

Exploratory Data Analysis revealed clear patterns in trip categories, payment types, and temporal trends. Our analysis reveals that most passengers took short trips (<2 miles) with 47.1M rides (Avg. Fare: $11.40), followed by Medium trips (2-10 miles) at 32.3M rides (Avg. Fare: $23.42). Even though Long trips (>10 miles) were taken the least at 6.8M rides, they have the highest unit revenue (Avg. Fare: $64.20) which makes sense because fares are charged by distance. 61.5M rides were paid for using credit cards, vastly outpacing cash (13.5M rides). Alternative or rare payment codes (3-5) appear in small numbers, which highlights a steep decline in cash usage and reliance on digital payment processing infrastructure. Average fares demonstrate clear seasonal demand elasticity. In 2024, average fares rose from $18.65 (Jan) to $20.55 (Dec), with peaks during winter holiday travel windows. Same in 2025, climbing from $18.33 (Jan) to $20.94 (Jun). Systemic clock anomalies (e.g., corrupted timestamps mapping data to 2002 or 2007) were identified as system noise and removed to ensure temporal trend integrity.

Analysis using the 7 Vs of Big Data

Volume: The dataset has millions of trip records per month, adding up to 1.85 billion rows overall. Even a single month is large enough to qualify as big data, with raw files exceeding 29 GB. This demonstrates the scale of information available for analysis. ***Velocity:*** Trips occur continuously, every hour of every day. While TLC publishes data monthly (introducing a delay), the underlying system generates data in real time. This highlights the difference between data generation velocity and publication velocity. ***Variety:*** Records include timestamps, numerical measures, categorical codes, geospatial coordinates, and financial values. This diversity allows analysis across multiple dimensions: time, space, behavior, and economics. ***Value:*** The dataset reveals demand patterns, high-volume pickup zones, revenue flows, and rider payment behavior. These insights can inform transport optimization, policy decisions, and business strategy. ***Veracity:*** Some fields (e.g., passenger counts) are self-reported and may be inaccurate. TLC itself warns that provider-submitted data may contain errors. Cleaning steps (removing nulls, duplicates, corrupted years) are essential to ensure trustworthy analysis. ***Variability:*** Demand fluctuates with season, weather, holidays, time of day, and special events. This variability affects fares, trip durations, and geographic distribution, making predictive modeling more complex. ***Visualisation:*** Tools such as line charts, histograms, scatterplots, boxplots, and heatmaps help identify patterns, relationships, and anomalies. Visualisation transforms raw data into interpretable insights.

Business and societal value

The yellow taxi dataset can help city planners, taxi companies, and the city of New York. City planners can use the records to identify areas in the city that have high taxi demand. Taxi companies can use the data to make their operations more efficient by mapping out where and at what time vehicles are most likely to find passengers. The city can evaluate transportation access, therefore these records can inform policy development.

The omission of cash tips introduces financial bias, which skews true driver earning profiles and trip profitability outcomes. The analysis of tips and payment types reveals how a customer behaves, which enables operators to refine pricing strategies and improve service quality. From the trip distance and pickup/dropoff zones we can possibly guide route planning, help reduce congestion, and improve fleet allocation. This would mean shorter wait times, lower emissions, and a better passenger experience. Though we derive some values from this data, we experience data quality issues such as corrupted timestamps (e.g., years outside 2022–2025), missing values, and self-reported passenger counts which reduce reliability. From a broader perspective, the dataset covers only NYC taxis, excluding ride-sharing platforms such as Uber, and other cities which limits its representativeness.

This is the link to the GitHub repository: https://github.com/michael-dlam/project

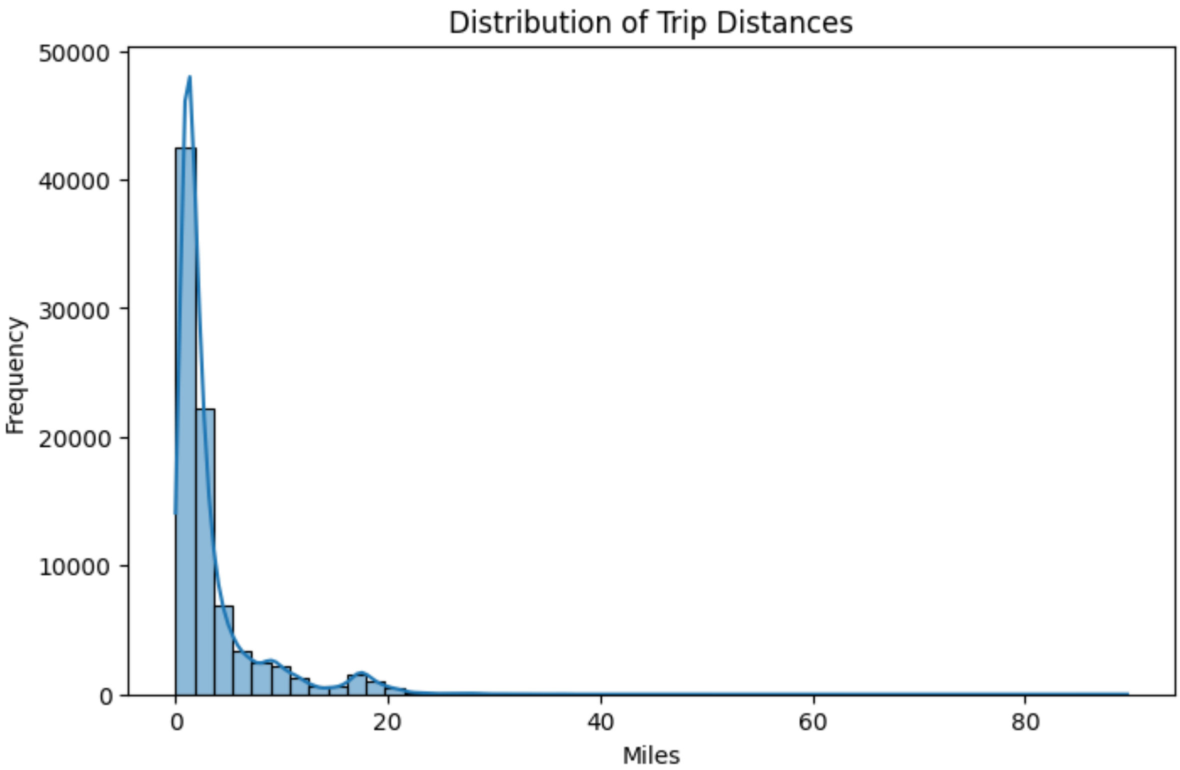
Appendix

trip_category	avg_fare	trip_count
long	64.20343481775933	6819043
medium	23.41773620736999	32336544
short	11.400070030667415	47114216

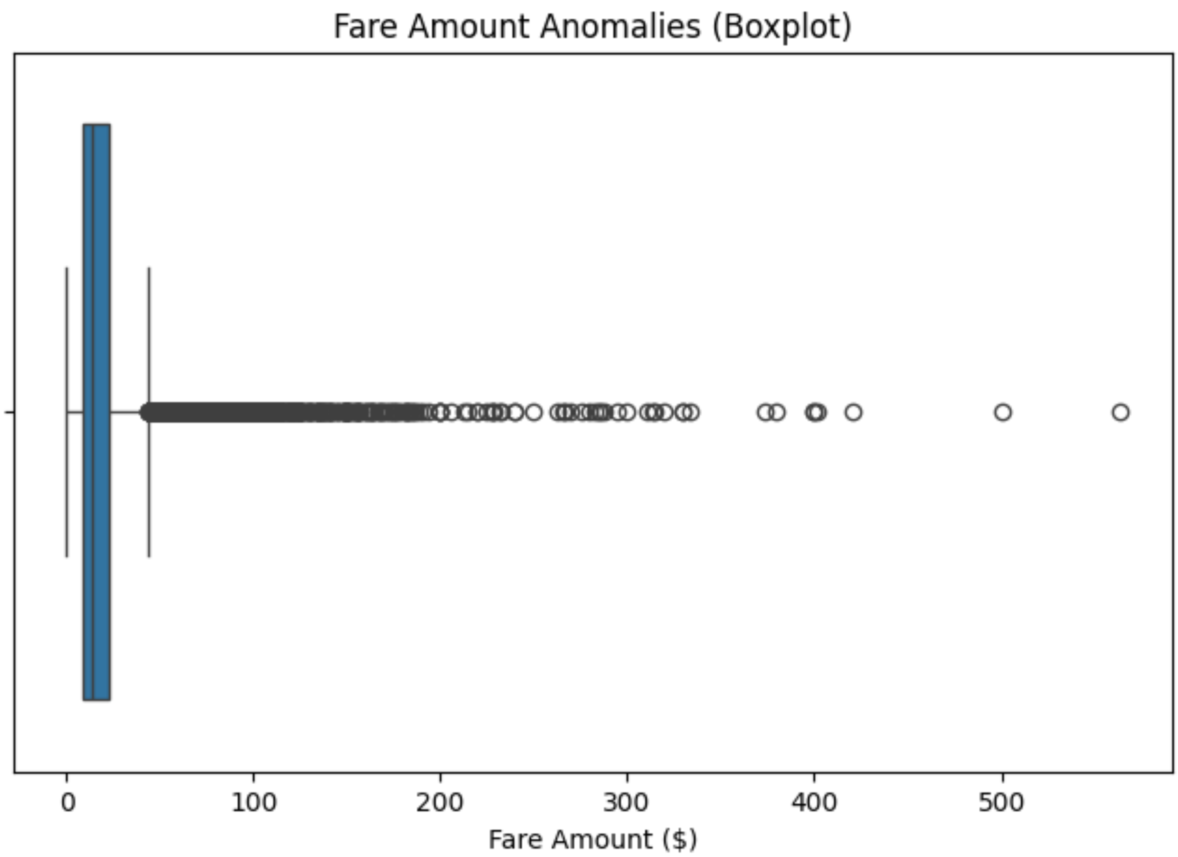
payment_type	count
1	61502227
0	13519338
2	9844100
4	999407
3	404729
5	2

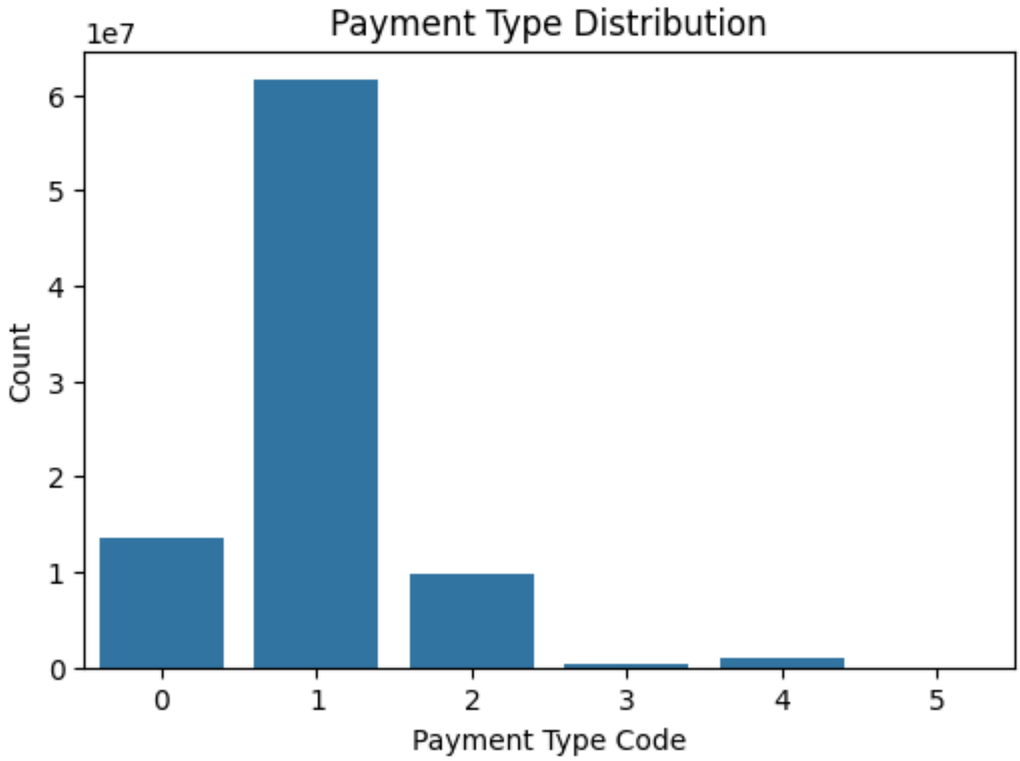
A numeric code signifying how the passenger paid for the trip.
0 = Flex Fare trip
1 = Credit card
2 = Cash
3 = No charge
4 = Dispute
5 = Unknown
6 = Voided trip

year	month	trip_count	avg_fare
2024	1	2926220	18.652230645679296
2024	2	2965965	18.544268462372187
2024	3	3522739	19.268825936295464
2024	4	3455276	19.572232536561334
2024	5	3661138	20.376806613134992
2024	6	3475970	20.088676513318514
2024	7	3017001	20.312077291323348
2024	8	2919660	20.552843944157825
2024	9	3558092	20.797751564602528
2024	10	3756781	20.502336833049306
2024	11	3573301	19.972494388242062
2024	12	3587753	20.552647763098516
2025	1	3329582	18.326291729712644
2025	2	3393302	18.16113569614497
2025	3	3934675	19.274663833734685
2025	4	3782127	19.423608879342233
2025	5	4264157	20.463793840142387
2025	6	4045205	20.937772189542937
2025	7	3650270	20.52487955685471
2025	8	3311241	20.329455944161104

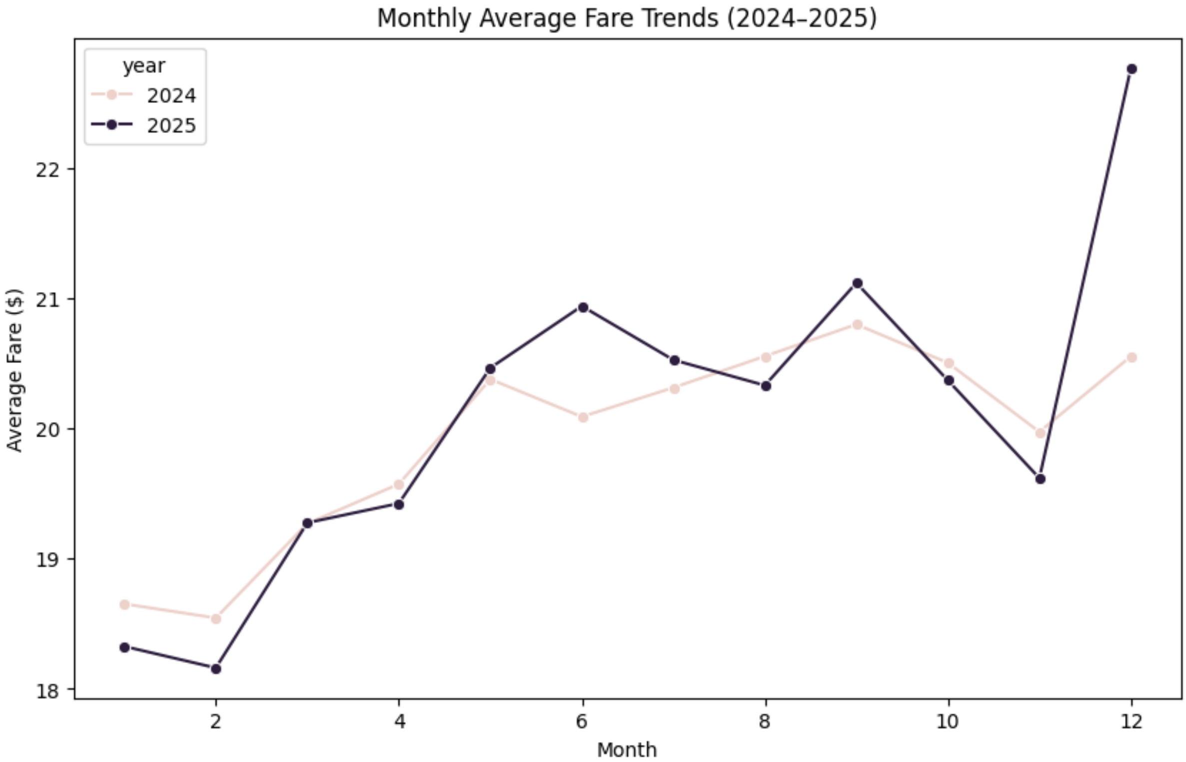

Distribution of Trip Distances
Frequency
Miles
0
10000
20000
30000
40000
50000
0
20
40
60
80


Trip Distance vs Fare Amount
Fare Amount ($)
Trip Distance (miles)
0
100
200
300
400
500
0
20
40
60
80


Fare Amount Anomalies (Boxplot)
0
100
200
300
400
500
Fare Amount ($)



A numeric code signifying how the passenger paid for the trip.
0 = Flex Fare trip
1 = Credit card
2 = Cash
3 = No charge
4 = Dispute
5 = Unknown
6 = Voided trip


Monthly Average Fare Trends (2024–2025)
year
2024
2025
Average Fare ($)
22
21
20
19
18
Month
2
4
6
8
10
12